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ANN vs CNN vs RNN: Neural Networks Guide



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September 30, 2024

Neural Networks can be challenging to understand. But if you look at them in the context of what they do and how they can help you, you can come to grips with them in no time.

While we are going to get a bit technical in this post, we have also written a post about specific [business applications of Neural Networks](#). From recognizing speech to [analyzing sentiment](#) in a data set and [classifying data](#), there's always a dedicated neural model that you can build — or use.

So in the guide, let's walk through:

- What exactly is a Neural Network is.
- The three major types and how they compare with each other, and
- A brief summary of other Neural Networks that you can use.

Let's dig in!

What is a Neural Network — and why should you care?

A Neural Network is a working system at the heart of a [Deep Learning](#) algorithm that helps it process raw data.

In fact, [Neural Networks](#) are pretty much like the human brain.

Like our brain that's formed of a network of neurons, Neural Networks are based on connected nodes or units — technically known as artificial neurons. Their job? To identify hidden correlations and patterns in raw data, classify them, and continuously improve.

But why bother learning about Neural Networks? As mentioned, neural networks sit at the center of all Deep Learning algorithms. So to understand a model or create one, you need to first dive into the architecture of a Neural Network.

With the basics out of the way, let's look at the three main types of Neural Networks.

Artificial Neural Networks (ANN)

Most folks consider Artificial Neural Networks synonymous with Neural Networks. However, both aren't one and the same. Instead, an ANN is a type of Neural Network.

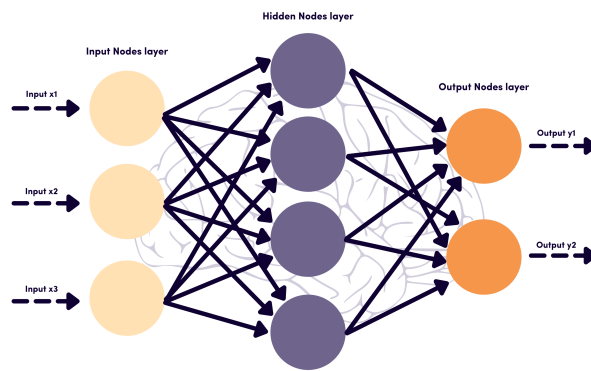
ANN is essentially a feed-forward network because of the way information travels through it — from one layer to another *without touching a node twice*.

This type of neural system — patterned around how neurons work in our brain — recognizes patterns in raw data, helping solve complex processes.

Another similarity with the human brain: the [ANN](#) improves with every new input it gets. In a nutshell, ANN makes independent improvements — or is continuously improving.

Now to dive into the composition of this Neural Network. The ANN is based on three or more interconnected layers of nodes — again, similar to the brain.

All layers are responsible for inputting, processing, and outputting data to the deeper layers. Such an interlayered system is what helps this Neural Network understand and learn complex things.



How Artificial Neural Networks work

What should you use ANNs for?

Owing to its tiered layering, the ANN is used in technology focusing on complex problem solving such as pattern recognition problems.

Examples include:

- Predictive analysis for business intelligence.
- Speech-to-text transcription application.
- Handwriting and facial recognition.
- [Spam email detection](#).
- Weather prediction.

Take a look at the more ways [AI is used in business](#).

Advantages of using an ANN

ANNs can bring many perks with their use, here are some of the main ones:

- **Adaptive nature.** Meaning: the Neural Network modifies itself post the initial training with subsequent information that it processes.
- **Enhanced learning ability.** Once trained, the ANN can learn events on its own. It can also make decisions or generate outcomes by comparing the problem with similar events. Due to their node strength, ANNs can perform multiple tasks at a time.
- **Gradual corruption.** A fault in one or a handful of cells in the ANN doesn't stop it from delivering output. Not only does this benefit make the Neural Network fault-tolerant but it also means that corruption tends to be slow. In fact, it occurs over time so you aren't left with an immediate problem on your hand.
- **Distributed information storage.** Unlike information storage in traditional programming, the ANN stores information on the entire network — not a specific database. This is another reason why it's fault-tolerant. A few pieces of missing information in one place don't bring the entire system down but are treated as outliers

Challenges of using an ANN

However, despite their many advantages, Artificial Neural Networks can also pose some challenges, which make it worth it to assess your company's needs and resources before deciding to train ANN:

- **It takes copious amounts of data to train an ANN.** The training consists of providing input to the network and telling it what the output should be.
- **Difficulty training the network.** There is no specific rule for determining the structure of Artificial Neural Networks — it often takes trial and error to get it right.
- **The architecture can be fuzzy.** When ANN gives a problem solution, it does not explicitly outline why and how this result was achieved.

Convolution Neural Networks (CNN)

Convolution Neural Networks are mainly credited for their role in image and video recognition, recommendation systems, and image analysis and classification.

Prior to it, manual labor was involved in identifying objects in images. CNN, however, has helped scale the process by using linear algebra principles to identify patterns in images.

So how do CNNs work? CNNs are based on three main layers. These are:

- Convolutional layer
- Pooling layer
- Fully-connected layer

With each layer, the CNN's complexity in understanding the image increases. This means the first layers focus on interpreting simple features in an image such as its edges and colors. As the image processes through layers, the network is able to

recognize complex features such as object shapes. Finally, the deepest layer is able to identify the target object.

What should you use CNNs for?

Broadly, this type of Neural Network is used for Computer Vision.

Essentially, Computer Vision is an AI field that's responsible for identifying meaningful information from visual inputs, images, and videos. For example, identifying dogs from a pile of images.

Such CNN-powered Computer Vision has several applications including:

- Recommend products that will likely complement an existing product or wardrobe.
- [Provide suggestions for tagging people in a photograph](#) as social media platforms do.
- Identify items as [Computer Vision](#) currently does for helping doctors better recognize cancerous tumors.

Advantages of using CNNs

CNNs show one main advantage:

- **It automatically detects features without human supervision.** This gives it an edge over its predecessors.

Challenges of using CNNs

As for its drawbacks, there are two main ones:

- **Difficulty in dealing with variance in the data presented.** CNN has a hard time processing objects in images that are hidden to an extent. With [image classification](#) too, the network has difficulty classifying tilted or rotated images. Put simply, CNN can't encode an object's orientation and position and can't process spatially invariant data.
- **Computationally demanding.** Training CNN requires numerous graphical processing units (GPUs). To add to that, if you lack good GPUs, the training becomes slow.

Recurrent Neural Networks (RNN)

RNNs are unique on account of their ability to process both past data and input data — and memorize things — and were developed to overcome the weaknesses of the feed-forward network. Practical applications include Google's voice search and Apple's Siri.

Like ANN and CNN, RNN also learns with training data. From there on, it doesn't process data on inputted data alone. Instead, it uses data from past inputs to make decisions too. In a nutshell, this architecture is built for having a 'memory'.

What should you use RNN for?

Because RNN has an internal memory, it can make relatively precise predictions. What's more, it's useful for solving *sequential* data-based problems.

Keeping this in mind, RNN applications include:

- Prediction problems
- Machine translation
- Speech recognition
- Sentiment Analysis
- Stock price forecasting
- Generating text and language modeling

Advantages of RNNs

Recurrent Neural Networks have unique abilities as opposed to other kinds of Neural Networks, which open a wide range of possibilities for their users — however, also bring some challenges with them. Here's a rundown of the main benefits:

- **It's the only Neural Network with memory and double data processing.**
- **It can map out several inputs and outputs.** Unlike other algorithms that deliver one output for one input, the benefit of RNN is that it can map out many to many, one to many, and many to one input and outputs.

Above all, RNNs have an in-depth understanding of sequences and their context in contrast with other Neural Networks.

Challenges of RNNs

With great benefits, naturally, come a few challenges:

- **Slow and complex training.** In comparison with other networks, RNN takes *a lot of time* in training. To add to that, the training is quite complex and difficult to implement.

- **Exploring or vanishing gradient concern.** Due to the way the layers are arranged in an RNN model, the sequence typically gets rather long. This results in the model training with exploding or null weights, leading to gradient concerns.
- **RNNs can't be stacked up.** For the RNN model to work, each node requires the output of the previous one. This means you can't stack up training or pair this model with another model. In turn, this leads to high computational expenses.

How should you evaluate your Neural Networks?

As with most things in [Machine Learning & Deep Learning](#), it mostly boils down to what you're trying to achieve — and what your success metrics are. You'll likely find structural and application differences in all NNs.

To help you see how each algorithm compares with another, we've summarized some of the parameters in the table below:

Basics	One of the simplest types of neural networks.	One of the most popular types of neural networks.	The most advanced and complex neural network.
Structural Layout	Its simplicity comes from its feed forward nature — information flows in one direction only.	Its structure is based on multiple layers of nodes including one or more convolutional layers.	Information flows in different directions, which gives it its memory and self-learning features.
Data Type	Feed on tabular and text data.	Relies on image data.	Trained with sequence data.
Complexity	Simple in contrast with the other two models.	Considered more powerful than the other two.	Fewer features than CNN but powerful due to its self-learning & memory potential.
Commendable Feature	Ability to work with incomplete knowledge and high fault tolerance.	Accuracy in recognizing images.	Memory and self-learning.
Feature type: spatial recognition	No	Yes	No
Feature type: Recurrent connections	No	No	Yes
Main Drawback	Hardware dependence.	Large training data required.	Slow and complex training and gradient concerns.
Uses	Complex problem solving such as predictive analysis.	Computer vision including image recognition	Natural language processing including sentiment analysis and speech recognition.

Table comparing the characteristics of CNN, ANN and RNN

Other types of NNs & when you should consider using them

Before we wrap this up, here's a quick rundown of other types of Neural Networks:

- **Perceptron.** This algorithm uses numerical inputs, weights, and a bias to make classifying predictions. Use it for data classification into two groups.
- **Multilayer perceptron.** This is a type of NN that has multiple layers of perceptrons. It's used for solving stochastic problems, giving you approximate solutions for complex problems, e.g. encoding databases or routinely checking database security consistency.
- **Feed-forward Neural Network.** Any Neural Network in which information flows in one direction is a FF Neural Network. Use it for weather prediction and other purposes that involve learning the relationship between independent variables.
- **Radial Basis Function (RBF) Neural Network.** RBF is a common type of feed-forward Neural Network based on three layers. It is, however, differentiated from other models due to its faster learning speed. Uses include classification, time series prediction, system control, and function approximation.
- **Modular Neural Network.** This is a type of ANN known for solving complex AI problems. Its structure is based on a set of Neural Networks — all independent (working on separate inputs and performing subtasks) and moderated using an intermediary.
- **Long Short-Term Memory (LSTM).** This is a type of RNN, named so because it uses short-term memory processes for creating long-term memory. It's used mainly in Deep Learning, e.g. for making predictions in speech recognition.
- **Sequence to sequence model (seq2seq).** This is another type of RNN in which either both or one of the input and output are sequences. It's involved in solving complex language problems. Consequently, its uses include creating chatbots, machine translation, and text summarization.

Get started

In short, you need different tooling to solve different problems. If you're just starting out on your [Machine Learning journey](#), it's important to know and define the problem you're trying to solve. It takes a great deal of effort to become proficient enough to build something on your own — and building literacy in this topic will help implementation processes.

Remember:

- ANNs (Artificial Neural Networks) are helpful for solving complex problems.
- CNNs (Convolution Neural Networks) are best for solving Computer Vision-related problems.
- RNNs (Recurrent Neural Networks) are proficient in [Natural Language Processing](#).

So... what are you building? Check out [how businesses use AI here](#).

Now that you're here

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